

Project Requirement Document

DOCUMENT NUMBER

PROJECT TITLE: Omah Kunci

CLIENT: Resource Manager Division





Table of Contents

Team Structure	3
Project Overview	4
1. Executive Summary	4
2. Design and Implementation Constraints	4
3. Assumptions and Dependencies	4
Project Requirements	5
4. Interfaces	5
5. Functional Requirements	5
6. Use-Case Requirements	5
7. Non-Functional Requirements	5
Project Execution	6
8. Project Scope	6
9. Project Deliverables	6
10. Timeframe and Milestones	6
11. Challenges and Open Questions	6



Team Structure

- Project Manager : Satya Wira Pramudita
- Client : Resource Manager Division
- Team
 1. Satya Wira Pramudita
 2. Ravif Gayuh Wicaksono
 3. Thufail Bahir Al Bariq
 4. Dhimas Putra Sulistio
 5. Hafidz Kurniawan Nahruntoko



Project Overview

1. Executive Summary

Omah Kunci is a key directory app where the key is the literal key to OmahTI's basecamp. Want to head to basecamp but don't have access to the keys? Omah Kunci lets you contact key holders! You can access it from the web or download the app on the Playstore or App Store.

2. Design and Implementation Constraints

Initially, the client requested an item tracking system to track the items in basecamp's inventory, though they restated that they had bigger issues that they were tackling, which was staff access to OmahTI's Basecamp. So we devised a platform for OmahTI's staff to access Basecamp without having to go through Resource Management first; they now can contact the key holders directly!

3. Assumptions and Dependencies

- Assumptions:
 1. Clients are willing to provide data on the key holders.
 2. Clients want quick and easy access to the product, so the UI must be simple and intuitive.
 3. OmahTI staff members shown on the website and application are willing to respond to key requests..
- Dependencies:
 1. The project depends on the willingness of OmahTI's staff who have access to the keys.
 2. User experience depends on the availability and response time of OmahTI staff members shown on the website.
 3. WhatsApp wa.me universal link and Hosting services are required to forward users to WhatsApp and to host the website.
 4. Synchronization for web development and app development is required to ensure product consistency.



Project Requirements

4. Interfaces

4.1. User Interfaces

The website and application will have a very simple and compact user interface so that users can intuitively navigate through and use our services. The themes and color scheme will use OmahTI's website as a reference. The main components of interactions are:

1. **Chat Buttons:**

Circular windows with pictures of key holders that, when you click or press them, will forward you straight to WhatsApp.

2. **Google Maps API:**

An interactive window that displays the location of OmahTI's basecamp through Google Maps API

4.2. Tech Stack

1. **Figma (UI)**

Using Figma, we will design the website and application.

2. **Next.js (with typescript)**

React-based framework for building fast, scalable web applications with built-in routing, SSR/SSG, and type safety.

3. **Tailwind CSS**

Utility-first CSS framework that allows for rapid and consistent styling directly within component markup.

4. **wa.me (WhatsApp chat link)**

Enables direct WhatsApp messaging through clickable links with pre-typed text messages, improving user interaction and communication.

5. **Vercel (hosting)**

Cloud platform for frontend frameworks, offering seamless deployment, automatic builds, and global CDN for high performance.



5. Functional Requirements

5.1. Basecamp Location

- **FR-1:** The system shall store and display details about OmahTI's basecamp using Google Maps API.
- **FR-2:** The location should be marked with a pin or marker.

5.2. Key Holder List

- **FR-3:** The system shall maintain a visible list of key holders associated with the basecamp.
- **FR-4:** Each key holder entry shall include name and clickable photo that opens WhatsApp

5.3. WhatsApp Integration

- **FR-5:** When a user clicks on one of the buttons, the browser shall open WhatsApp with a pre-filled message.

5.4. Frontend Behavior

- **FR-6:** The layout shall be responsive and accessible on both desktop and mobile devices.

6. Non-Functional Requirements

- **Web Performance**

The website must be optimized for fast loading times to provide a seamless and responsive user experience. Efficient performance is essential to maintain user engagement and reduce bounce rates.

- **Availability**

After deployment, the website should remain consistently accessible to users. Downtime should be minimal, occurring only during scheduled maintenance or in the case of unexpected technical issues.

- **Usability**

The website should feature a user-friendly interface with intuitive navigation. All elements must be easy to understand and operate, ensuring that users can interact with the system efficiently without the need for guidance.

- **Maintainability**

The source code should be structured clearly, using standardized formatting, meaningful comments, and a logical file organization. This ensures that future developers can easily read, troubleshoot, or enhance the codebase as needed.



Project Execution

7. Project Scope

We are designing a simple and responsive website that focuses on two main features: a contact information panel and a location panel. The first section of the website will include a brief introduction and description of our project, OmahKunci. The second section will display images of individuals who have access to the basecamp key. When a user clicks on one of these images, they will be redirected to a personal messaging platform to contact the respective individual directly. Finally, the website will include a map showing the exact location of the OmahTI basecamp, which will be helpful for users or clients who are unfamiliar with its precise location.

In addition to the website, we are also developing a mobile application version of OmahKunci. This mobile app is designed to provide the same core functionality while ensuring users can access the platform anytime and anywhere, directly from their smartphones. The mobile version aims to enhance usability, especially for users who are frequently on the move.

8. Project Deliverables

As a result of the scope mentioned above, OmahTI will deliver:

- **Fully functional website,**

A responsive and user-friendly website that includes all planned features: introduction section, contact panel with clickable profiles, and an interactive location map.

- **All source code files,**

All website files including HTML, CSS, JavaScript, images, and any relevant assets or documentation.

- **Design assets,**

All design-related resources used in the development of the website, including UI mockups, icons, images, and font files



- **A link to the deployed website,**

Accessible URL where the website can be viewed and tested. The site will be hosted on Vercel.

- **App Installation Package,**

A ready-to-install app package for Android or iOS, enabling easy distribution for testing or usage among the target users.

- **Final presentation,**

A presentation summarizing the project goals, design process, key features, and a live demo of the website.on.

9. Timeframe and Milestones

Date	Milestones	Description
Week 1	Logo making	Designing a logo that represents the OmahKunci project. This includes selecting appropriate colors and visual elements.
Week 2	UI design	Creating the user interface layout for both the website and mobile app. This includes designing mockups or wireframes for key sections such as the homepage, contact panel, and map section.
Week 3	Idea implementation	Starting the development process by converting designs into functional code. This includes building the basic structure, integrating clickable contact elements, and embedding the location map.



10. Challenges and Open Questions

- Data collection

The time required to collect data from the resource manager took quite a long time so the implementation process was delayed.



Gedung Fakultas MIPA UGM, Sekip 
Utara, Bulaksumur, Sinduandi, Mlati,
Sleman, D.I Yogyakarta

omahti.mipa@ugm.ac.id 

Visit us on
omahti.web.id